

Branch: gh-pages **genstats / inst / doc / 02\_05\_normalization.R**

jtleek 12 days ago big update

1 contributor

71 lines (58 sloc) 2.51 KB

Raw

Blame

History



```
1  ## ----global_palette, results = 'asis'-----
2  rm(list=ls())
3  tropical= c('darkorange', 'dodgerblue', 'hotpink', 'limegreen', 'yellow')
4  palette(tropical)
5
6  ## ----global_options,warning=FALSE,message=FALSE-----
7  ## see ch. 10 Hooks of Xie's knitr book
8  library(knitr)
9  knit_hooks$set(setPch = function(before, options, envir) {
10    if(before) par(pch = 19)
11  })
12  opts_chunk$set(setPch = TRUE)
13
14  ## ----global_plot,warning=FALSE, message=FALSE-----
15  knitr::opts_chunk$set(fig.width=5, fig.height=5, size="footnotesize",
16                        warning=FALSE, message=FALSE)
17  knitr::knit_hooks$set(small.mar = function(before, options, envir) {
18    if (before) graphics::par(mar = c(5,5,1.5,1))
19  })
20
21  ## ----load_hidden, echo=FALSE, results="hide", warning=FALSE-----
22  suppressPackageStartupMessages({
23    library(devtools)
24    library(Biobase)
25    library(preprocessCore)
26  })
27
28  ## ----load-----
29  library(devtools)
30  library(Biobase)
31  library(preprocessCore)
32
33  ## ----install_packages, eval=FALSE-----
34  # install.packages(c("devtools"))
35  # source("http://www.bioconductor.org/biocLite.R")
36  # biocLite(c("Biobase","preprocessCore"))
37
38  ## -----
39  con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/montpick_eset.RData")
40  load(file=con)
41  close(con)
42  mp = montpick.eset
43  pdata=pData(mp)
44  edata=as.data.frame(exprs(mp))
45  fdata = fData(mp)
46  ls()
47
48  ## -----
49  edata = log2(edata + 1)
50  edata = edata[rowMeans(edata) > 3, ]
51  colramp = colorRampPalette(c(3,"white",2))(20)
52  plot(density(edata[,1]),col=colramp[1],lwd=3,ylim=c(0,.30))
53  for(i in 2:20){lines(density(edata[,i]),lwd=3,col=colramp[i])}
54
55  ## -----
56  norm_edata = normalize.quantiles(as.matrix(edata))
57  plot(density(norm_edata[,1]),col=colramp[1],lwd=3,ylim=c(0,.20))
58  for(i in 2:20){lines(density(norm_edata[,i]),lwd=3,col=colramp[i])}
59
60  ## -----
61  plot(norm_edata[,1],col=as.numeric(pdata$study))
62
63  ## -----
```

```
64  svd1 = svd(norm_edata - rowMeans(norm_edata))
65  plot(svd1$v[,1],svd1$v[,2],xlab="PC1",ylab="PC2",
66       col=as.numeric(pdata$study))
67
68  ## ----session_info-----
69  devtools::session_info()
70
```

